Modeling Carryover from Pharmacodynamic Principles: A Generative Proposal for the Mean, Moderation and Covariance of the Drug Response

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Purpose. This paper proposes, from pharmacodynamic principles, a generative model of the drug response in which carryover is specified once and the mean, the biomarker moderation and the covariance of the response all follow from it. It sets out the model, derives its implied moments and the consequences for the interaction test, and compares it with the constructs in use. It is a proposal: its results are derived, not implemented or simulated.

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Notation index and glossary

The notation follows `analysis/report/NOTATION.md`. Symbols marked *canonical* carry the meaning given there; symbols marked *local* are introduced here and chosen not to collide with canonical symbols. The proposal's sensitivity, moderation-strength and noise symbols were renamed on 2026-10-07 (θ, κ, u, ε, γ in the first draft) because those letters carry other canonical meanings. On 2026-10-09 the contrast-biomarker correlation was renamed from $\rho_{\Delta B}$ to $r_{\Delta B}$, as in `docs/37` and `analysis/report/compendium-mathematics.tex` (ρ is the canonical serial correlation), and the normal law written $\mathcal{N}$.

Table 1: Notation index for the pharmacodynamic carryover model

Symbol	Meaning	Status
i, t, s	patient; visit (two visits t, s)	canonical
B_i, b_i	biomarker; standardized biomarker $(B_i - \mu_{bm})/\sigma_{bm}$	canonical
μ_{bm}, σ_{bm}	biomarker mean and standard deviation	canonical (σ_{bm})
BR_{it}	drug (biological) response component	canonical
Y	symptom score, the outcome	canonical
c_{bm}	covariance-moderation strength of the published construct	canonical
D_{bc}	exposure-decayed continuous drug indicator	canonical
$\beta_{bm:D}$	interaction slope: population slope of the contrast Δ on B	canonical
t_{sd}, $t_{1/2}$	time since discontinuation; carryover half-life as used by the published construct	canonical
$t^R_{1/2}$	response (pharmacodynamic offset) half-life	local
k_{out}	first-order turnover rate of the response, $\ln 2/t^R_{1/2}$	local
$f(t)$	population drug-response time course, scaled to 1 at full on-drug effect	local
$f^*(t)$	drug-driven target of the turnover model: 1 on drug, 0 off drug	local
t_{stop}	time of discontinuation	local
ω_i, $\bar{\omega}$	patient i's drug sensitivity and its population mean	local
ζ	strength of multiplicative moderation of sensitivity by b_i	local
v_i, σ_v	sensitivity variation not explained by the biomarker, and its SD	local
ξ_{it}, σ_ξ, $r_\xi(t,s)$	serial within-patient noise of BR, its SD and its correlation	local
c^{peak}_{bm}	biomarker-response correlation at full effect, a derived value	local

Symbol	Meaning	Status
a_t	E9 contrast weights, $1/n_{\text{on}}$ on drug, $-1/n_{\text{off}}$ off drug	local (as `docs/37`)
Δ	patient's on-minus-off contrast (E9)	local (as `docs/37`)
F	contrast-weighted exposure, $\bar{f}_{\text{on}} - \bar{f}_{\text{off}}$	local
$\bar{f}_{\text{on}}$, $\bar{f}_{\text{off}}$	mean of $f(t)$ over on-drug and off-drug visits	local
V_0	contrast variance from serial noise and the other response factors	local
$r_{\Delta B}$	correlation between contrast and biomarker	local (as `docs/37`)
BL, TV, PB	baseline level; natural-course and placebo-belief components	canonical
n_{on}, n_{off}	on-drug and off-drug visits of a path	local (as `docs/37`)
G, H	shorthand $\bar{\omega}\zeta$ and $\bar{\omega}^2\zeta^2 + \sigma_v^2$ in the proof of Section 6	local

Glossary.

- **Pharmacokinetic carryover.** Drug remaining in the body after the last dose.
- **Pharmacodynamic (disease-state) carryover.** Improvement that persists after the drug has been eliminated.
- **Turnover (indirect-response) model.** A response that relaxes toward a drug-driven target at a first-order rate.
- **Anchored decay.** Off-drug decay measured from the level reached at discontinuation.
- **Moderation profile.** The fraction of the full biomarker moderation present at a visit (`docs/37`).
- **Step rule, graded coupling, decayed mean moderation.** The coupling rules of `docs/37`, Section 2.
- **E9.** The paired-difference interaction statistic of `docs/37`, Section 3.
- **`orig`, `covar`, configurations C and D.** The published (`58b32a9`) construct, the package's AR(1) construct, and the configurations of `docs/36`, Appendix A.6.

1. Summary

`docs/36` and `docs/37` traced both problems of the Hendrickson et al. (2020) simulation framework, invalid correlation matrices and the collapse of power

under small carryover, to the way the framework specifies its correlation matrix: entry by entry, with the carryover half-life acting on the off-drug mean through one formula and on the biomarker correlation through a separate rule. This paper argues, from pharmacodynamic principles, that carryover should not be specified separately for the mean and for each element of the correlation matrix. It should be specified once, in a generative model of each patient's drug response, from which the mean, the variance, the biomarker correlation and the cross-visit correlations all follow.

The proposal:

1. **Carryover is pharmacodynamic.** For the prazosin and posttraumatic stress disorder (PTSD) context, the drug leaves plasma with a half-life of approximately 2.3 hours (Hobbs et al., 1978), so pharmacokinetic carryover is essentially complete within a day. Persistence of benefit beyond that is a property of the response, and the half-life parameter should be the response (offset) half-life (Section 2).
2. **The mean follows a turnover model.** The response relaxes toward a drug-driven target at a first-order rate (Dayneka et al., 1993). Off drug this gives decay anchored at the level reached at discontinuation, and on restart the response builds from its residual level (Section 3).
3. **Moderation scales with the response.** A predictive biomarker acts on drug sensitivity, so the biomarker-dependent part of the response rises and decays with the response itself (Section 4).
4. **The covariance is implied, not specified.** Writing each patient's drug response as a random sensitivity times the population time course, plus serial noise, makes the covariance a sum of positive semidefinite terms. It is valid for every design and half-life by construction, the biomarker correlation follows the response, and the drug-response variance shrinks as the effect washes out (Section 5).
5. **The closed-form results of `docs/37` extend.** The interaction slope is proportional to the contrast-weighted exposure $F = \bar{f}_{\text{on}} - \bar{f}_{\text{off}}$, and power is strictly increasing in F; unexplained variation in drug sensitivity caps the attainable power (Section 6).

The paper is a proposal. Sections 3 to 6 are derivations, not simulation results; nothing in it has been implemented or simulated.

2. Two kinds of carryover

Pharmacokinetic carryover is drug remaining in the body after the last dose. For prazosin it is short: in 24 normotensive subjects, drug left plasma with a half-life of approximately 2.3 hours (Hobbs et al., 1978). Five half-lives, about 12 hours, remove 97% of it. Hendrickson et al.'s half-lives of 0.1 and 0.2 weeks (17 and 34 hours) are on this scale or slightly above it.

Pharmacodynamic, or disease-state, carryover is improvement that persists after the drug has gone: consolidated sleep, fewer nightmares, behavioral learning, recovery of a physiological set point. Its time course is set by the turnover of the response, not by drug elimination, and when the response is slow to turn over it can last days to weeks. Effect-compartment models (Sheiner et al., 1979) and indirect-response models (Dayneka et al., 1993) are the standard quantitative treatments of this delay between concentration and effect; in indirect-response models in particular, the response returns to baseline slowly after the drug is gone.

For a symptom outcome measured weekly or less often, pharmacokinetic carryover is negligible and the carryover that matters is pharmacodynamic. The half-life parameter of the simulation should therefore be read, and chosen, as the response half-life $t_{1/2}^{R}$. The appropriate value for prazosin in PTSD is not known to us; it is an empirical quantity that the simulation should treat as uncertain.

3. The mean: a turnover model

Let $f(t) \in [0, 1]$ denote the population drug-response time course, scaled so that $f = 1$ at the full on-drug effect. A minimal turnover model is

$$\frac{df}{dt} = k_{\text{out}}\,(f^{*}(t) - f(t)), \qquad k_{\text{out}} = \frac{\ln 2}{t_{1/2}^{R}},$$

where $f^{*}(t)$ is the drug-driven target: the full on-drug effect (1 on this scale) while on drug, 0 off drug. For a target that is constant within each on-drug or off-drug interval, the solution is exact and piecewise: off drug,

$$f(t) = f(t_{\text{stop}})\; 2^{-t_{sd}/t_{1/2}^{R}},$$

decaying from the level reached at discontinuation; on drug, f approaches the target from wherever it started. Three properties follow, each of which differs from the published construct:

- **Anchored decay.** Off-drug decay is measured from the level at discontinuation. The published code multiplies the previous visit's already-decayed mean by $2^{-t_{sd}/t_{1/2}}$ with t_{sd} cumulative, which decays faster than the stated half-life (`docs/36`; for off-drug visits one and two weeks after stopping, the second is at $2^{-3/t_{1/2}}$ rather than $2^{-2/t_{1/2}}$).
- **Restart from the residual.** When the drug is restarted the response builds from its residual level. The published code restarts the Gompertz onset curve from zero, as if the earlier course had not occurred.

- **One mechanism.** Onset and offset come from the same process. If a Gompertz onset is retained for fidelity to the published trajectories, the offset should at least be anchored and the restart should begin from the residual.

The simplest change compatible with the published model is to keep the Gompertz onset, anchor the offset, and start each new on-drug interval from the residual level. The full turnover model is the principled version.

4. Moderation scales with the response

A predictive biomarker modifies the drug's effect: in pharmacological terms it acts on sensitivity, the maximal effect or the potency of the drug in that patient. Write patient i's drug response as

$$BR_{it} = \omega_i\, f(t) + \xi_{it}, \qquad \omega_i = \bar{\omega}\,(1 + \zeta b_i) + v_i,$$

with b_i the standardized biomarker, ζ the strength of moderation, $v_i \sim \mathcal{N}(0, \sigma_v^2)$ the variation in sensitivity that the biomarker does not explain, and ξ_{it} serial within-patient noise with variance σ_ξ^2, independent of b_i and v_i. The mean response is $\bar{\omega} f(t)$, and the biomarker-dependent part, $\bar{\omega}\zeta b_i f(t)$, has the same time course as the response.

The *moderation profile*, the fraction of the full moderation present at visit t, is therefore $f(t)$ itself:

- **On drug**, moderation grows with the response. It is not at full strength from the first on-drug visit; it is small early in titration, when the response is small.
- **Off drug**, moderation decays with the response. It neither holds at full strength while the drug effect washes out, as under the published step rule (`docs/37`, Section 5.1), nor stops the moment the drug does, as under the current mean-moderation construct (configuration D of `docs/36`), which shifts the response at on-drug visits only.

The decayed mean moderation of `docs/37` (Section 5.6), with an anchored profile equal to 1 on drug and $2^{-t_{sd}/t_{1/2}}$ off drug, is an approximation to this; the proportional profile $f(t)$ is the form the pharmacology implies.

5. The implied covariance

Under the model of Section 4 every second moment of the drug response follows from four quantities, $\bar{\omega}\zeta$, σ_v, σ_ξ and the serial correlation r_ξ of ξ (derived):

$$
\begin{aligned}
\mathrm{Var}(BR_t) &= (\bar{\omega}^2\zeta^2 + \sigma_v^2)\, f(t)^2 + \sigma_\xi^2, \\
\mathrm{Cov}(BR_t, BR_s) &= (\bar{\omega}^2\zeta^2 + \sigma_v^2)\, f(t) f(s) + \sigma_\xi^2\, r_\xi(t,s), \\
\mathrm{Cov}(B, BR_t) &= \sigma_{bm}\, \bar{\omega}\zeta\, f(t), \qquad \mathrm{Cor}(B, BR_t) = \frac{\bar{\omega}\zeta\, f(t)}{\sqrt{(\bar{\omega}^2\zeta^2 + \sigma_v^2) f(t)^2 + \sigma_\xi^2}}.
\end{aligned}
$$

Four consequences:

- **The biomarker correlation follows the response.** It is zero where $f = 0$, rises and decays with f, and is never a step. The quantity the published framework calls c_{bm} becomes a derived value, most naturally the correlation at full effect, $c_{bm}^{\text{peak}} = \bar{\omega}\zeta / \sqrt{\bar{\omega}^2\zeta^2 + \sigma_v^2 + \sigma_\xi^2}$.
- **The drug-response variance shrinks off drug.** As the effect washes out, the sensitivity term vanishes and only the serial noise remains. The published construct gives the drug-response factor its full standard deviation (8) at every visit, including off-drug visits with zero mean, as if patients had a full-sized random drug response with no drug. Of the construct's features this is, in our judgment, the least defensible, and it adds noise to every on-off contrast.
- **Cross-visit correlation has structure.** It is a rank-one sensitivity term, $f(t)f(s)$, plus the serial correlation of the noise. For symptom data a decaying serial correlation (AR(1) or continuous-time AR(1)) is more plausible than compound symmetry.
- **Validity by construction.** The covariance of the drug responses is a sum of a positive semidefinite rank-one term and a positive definite noise covariance, so it is positive definite for every design, every schedule and every half-life. There is no ceiling on c_{bm} to exceed and nothing to repair: an impossible combination of correlations cannot be written down, because correlations are outputs rather than inputs.

The other response factors follow the same logic. The time-related (natural history) and expectancy (placebo) factors are separate processes; correlation between them and the drug response should arise from shared patient-level effects, such as a patient's general responsiveness entering both ω_i and the expectancy effect, rather than from constants placed in matrix cells. The resulting covariance is again a sum of positive semidefinite terms, in the form of the linear model of coregionalization of `docs/35`.

Data from this model need no correlation matrix at all: draw b_i, v_i and the serial noise, compute $f(t)$ for the patient's schedule, and assemble BR_{it}. The construction is the factored generator of `docs/36`, Section 6.7, without the matrix.

6. Consequences for the interaction test

The closed-form approach of `docs/37` extends directly (derived; not verified by simulation). With the paired-difference statistic E9 and contrast weights a_t ($1/n_{\text{on}}$ on drug, $-1/n_{\text{off}}$ off drug), define the contrast-weighted exposure

$$F = \sum_t a_t f(t) = \bar{f}_{\text{on}} - \bar{f}_{\text{off}}.$$

The drug-response part of the sum $\sum_t a_t BR_{it}$ is $\omega_i F + \sum_t a_t \xi_{it}$; because $Y = \text{BL} - (TV + PB + BR)$, it enters the contrast $\Delta_i = \sum_t a_t Y_{it}$ with a negative sign. Writing V_0 for the variance contributed by the serial noise and the other response factors, assumed independent of ω_i,

$$\beta_{bm:D} = -\frac{\bar{\omega}\zeta\, F}{\sigma_{bm}}, \qquad \text{Var}(\Delta) = (\bar{\omega}^2\zeta^2 + \sigma_v^2)\, F^2 + V_0, \qquad r_{\Delta B} = -\frac{\bar{\omega}\zeta\, F}{\sqrt{(\bar{\omega}^2\zeta^2 + \sigma_v^2)F^2 + V_0}}.$$

(The first draft gave $r_{\Delta B}$ without the minus sign; the sign follows from that of $\beta_{bm:D}$ and does not affect power, which depends on $|r_{\Delta B}|$. Corrected 2026-10-07.)

Proposition. *$|r_{\Delta B}|$ is strictly increasing in $|F|$.* With $G = \bar{\omega}\zeta > 0$ and $H = \bar{\omega}^2\zeta^2 + \sigma_v^2$, $|r_{\Delta B}| = G|F|/\sqrt{HF^2 + V_0}$, whose derivative in $|F|$ is $GV_0/(HF^2 + V_0)^{3/2} > 0$ for $V_0 > 0$. Since power is increasing in $|r_{\Delta B}|$ (`docs/37`, Section 4.4), power is strictly increasing in the magnitude of the contrast-weighted exposure (derived).

The half-life enters through F. Under anchored offset with the on-drug course held fixed, each off-drug $f(t)$ increases with $t_{1/2}^R$, so F decreases and power falls monotonically, as in `docs/37`, Proposition 3. Under the full turnover model a longer response half-life also slows onset and changes the restart level, so F is not guaranteed to be monotone in $t_{1/2}^R$ for every schedule; it must be checked per design. The monotone relation between power and F holds regardless.

Two features differ from the published construct:

- **Variance depends on carryover.** In `docs/37`, Proposition 1, the contrast variance did not depend on the half-life. Here the term $(\bar{\omega}^2\zeta^2 + \sigma_v^2)F^2$ does, because sensitivity variation contributes noise in proportion to the exposure contrast.
- **Power is capped by what the biomarker explains.** The bound $|r_{\Delta B}| < \bar{\omega}\zeta/\sqrt{\bar{\omega}^2\zeta^2 + \sigma_v^2}$ holds for every F and is approached as HF^2 dominates V_0; it is the square root of the share of sensitivity variation the biomarker accounts for. Since $f \in [0, 1]$ gives $|F| \leq 1$, the attainable value is lower still. No design and no amount of exposure contrast can

exceed the bound. This is a property of the biology the model encodes, not of the simulation's construction, and it gives the effect size a direct interpretation.

7. Comparison with the constructs in use

Table 2: Features of orig, covar, graded separable, mean moderation and proposed constructs

Feature	`orig` (58b32a9)	`covar` (package)	Graded + separable (`docs/36` C)	Mean moderation (D)	Proposed
Off-drug mean	recursive decay	anchored decay	recursive (as published)	recursive (as published)	anchored, from turnover
Restart after off-drug interval	from zero	from zero	from zero	from zero	from residual level
Moderation on drug	full at every visit	full	full	full	proportional to response
Moderation off drug	full if any residual mean (step)	decays	decays	none	proportional to response
Drug-response variance off drug	full	full	full	full	shrinks with response
Valid for every schedule	no (ceiling, repair)	no (ceiling, Cholesky fallback)	response block yes; ceiling remains	yes	yes, by construction
Ceiling on effect size	yes	yes	yes	none	none (correlations are outputs)

The mean and restart entries for `covar` were inspected in the package source: `R/generateData.R` decays the off-drug mean from the last on-drug value, and `R/buildtrialdesign.R` resets time on drug to zero after an off-drug interval. The remaining `covar` entries summarize `docs/32` and `docs/36`.

8. Implications for analysis and design

- **Analysis.** The analysis should use an exposure variable built from the same time course: the continuous indicator D_{bc} of the compendium, with the response half-life rather than the drug half-life, weights each visit by its expected exposure. A nonlinear mixed model of the turnover process, fitted directly, is the principled alternative and would estimate the response half-life.
- **Design.** Off-drug visits early after discontinuation carry the least contrast when the response half-life is long. Washout or off-drug assessment at four to five response half-lives restores $\bar{f}_{\text{off}} \approx 0$; if the response half-life is a week or more, an assessment one week after stopping contributes little to the interaction whatever the analysis.
- **Sensitivity.** Because the response half-life is uncertain, power should be reported across a range of plausible values rather than at a single assumed one.

9. What adopting the model would involve

- **Package.** A new data-generating option alongside `dgp_architecture = 'mvn'` and `'mean_moderation'` in `R/generateData.R`, with parameters for sensitivity ($\bar{\omega}$, matching the Gompertz maximum), moderation (ζ or c_{bm}^{peak}), unexplained sensitivity variation (σ_v), serial noise (σ_ξ and its correlation) and the response half-life. No correlation matrix is formed.
- **Checks.** With $\zeta = 0$ the interaction test must hold its size; with $t_{1/2}^{R} \to 0$ the results must reduce to the no-carryover case; the simulated moments must match Section 5.
- **Compendium.** Every paper that simulates through `generateData()` is affected in its data-generating assumptions, and results would need to be compared with the current architectures before any are replaced. The closed-form results of `docs/37` carry over with the coupling gap replaced by the contrast-weighted exposure F.

This is a substantial change to the data-generating process and is offered as a proposal for decision, not as a completed revision.

10. Limitations and evidence status

- **Derived, not tested.** Sections 3 to 6 are derivations. No part of the proposed model has been implemented or simulated.
- **Pharmacological premises.** The prazosin plasma half-life is from one study in 24 normotensive volunteers (Hobbs et al., 1978). The claim

that persistent benefit in PTSD is pharmacodynamic rather than pharmacokinetic follows from that half-life and from general pharmacodynamic principles; the response half-life itself is unknown to us.

- **Multiplicative moderation is a modeling choice.** It assumes the biomarker acts on sensitivity. A biomarker that acted on the onset or offset rate would require moderation of k_{out} instead, and would give a different moderation profile.
- **Normality.** The derivations of Section 6 assume joint normality of b_i, v_i and the noise; with a multiplicative random effect ($\omega_i f(t)$, with f a fixed function of the schedule), the responses are normal for each schedule, but extensions with a nonlinear link would not be.
- **Simplifications.** The other response factors are assumed independent of ω_i in Section 6; shared patient-level effects would add covariance terms to V_0 and to $\beta_{bm:D}$.

11. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020; 2:13. doi:10.3389/fdgth.2020.00013. Code: `github.com/rchendrickson/pmsimstats`, commit `58b32a9`.
2. Hobbs DC, Twomey TM, Palmer RF. Pharmacokinetics of prazosin in man. *Journal of Clinical Pharmacology* 1978; 18(8-9):402-406. doi:10.1002/j.1552-4604.1978.tb02456.x (PMID 690251).
3. Dayneka NL, Garg V, Jusko WJ. Comparison of four basic models of indirect pharmacodynamic responses. *Journal of Pharmacokinetics and Biopharmaceutics* 1993; 21(4):457-478. doi:10.1007/BF01061691 (PMID 8133465).
4. Sheiner LB, Stanski DR, Vozeh S, Miller RD, Ham J. Simultaneous modeling of pharmacokinetics and pharmacodynamics: application to d-tubocurarine. *Clinical Pharmacology and Therapeutics* 1979; 25(3):358-371. doi:10.1002/cpt1979253358 (PMID 761446).
5. pmsimstats team. `docs/35-separable-response-covariance.md`, `docs/36-hendrickson-pd-and-carryover.md`, `docs/37-carryover-power-closed-form.md`.